Pamantasan ng Lungsod ng Maynila

ENHANCEMENT OF THE GIRVAN-NEWMAN ALGORITHM APPLIED IN URBAN INFRASTRUCTURE FOR OPTIMIZED WATER NETWORK SECTORIZATION

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APPROVAL SHEET

The thesis hereto titled

ENHANCEMENT OF THE GIRVAN-NEWMAN ALGORITHM APPLIED IN URBAN INFRASTRUCTURE FOR OPTIMIZED WATER NETWORK SECTORIZATION

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Chapter ONE
INTRODUCTION

1.1 Background of the Study

The strategic management of urban water systems has become a central pillar of the global smart city movement, as municipalities look to modernize aging infrastructure through data-driven approaches. One of the most prominent trends in this sector is Water Network Sectorization (WNS), a process defined by the partitioning of a complex water distribution network (WDN) into smaller, manageable zones known as District Metered Areas (DMAs). According to a review by MDPI (2020), this "divide and conquer" methodology is essential for the reduction of non-revenue water, the simplification of pressure management, and the rapid localization of pipe bursts. By creating these sectors, utilities can transform a massive, opaque system into a series of transparent, independently monitored clusters.

However, the utility of the Girvan-Newman (GN) Algorithm as a baseline for sectorization is severely hindered by both physical and computational limitations. While Jazayeri and Moeini (2024) and Tian et al. (2022) note its "physical blindness" to hydraulic parameters, often resulting in severe pressure drops and hydraulically unstable sectors, the algorithm also suffers from three fundamental structural flaws. Specifically, continuous path recalculations impose an unfeasible O(n^3) time complexity on large networks. Furthermore, its greedy, top-down approach frequently and irreversibly fragments graphs into meaningless single-node communities. Finally, the algorithm's arbitrary tie-breaking method for edges with identical betweenness scores yields unstable and irreproducible partitions.

The primary objective of this study is to develop an enhanced Girvan-Newman (GN) algorithm specifically optimized for urban infrastructure sectorization. To evaluate

the effectiveness of the proposed approach, the study will conduct a series of comparative analyses. These analyses will benchmark the computational efficiency, partition stability, and hydraulic reliability of the enhanced variant against the standard GN model using established reference networks, such as C-Town and L-Town. Ultimately, the study seeks to provide water utilities with a fast, stable, and physically aware tool to achieve optimal network sectorization.

The contribution of this research lies in its ability to bridge the gap between abstract graph theory and the physical laws of hydraulics. As highlighted by Saldarriaga et al. (2025), the integration of hydraulic parameters into graph-based partitioning is critical for the long-term resilience of urban utilities. This study provides a scalable, computationally efficient framework that allows for the creation of DMAs that respect both topological modularity and physical pressure constraints, ultimately contributing to more sustainable urban water management.

Existing Algorithm:

Step 0: Start

Step 1: Network Graph Initialization

a. Represent the water network as an undirected graph G = (V, E)

b. Assign unit weights to all edges E (pipes), treating them as equal links.

Problem 2

Step 2: Calculate Global Edge Betweenness

a. Compute the shortest paths between all pairs of nodes (s, t) in the entire network.

b. For every edge, calculate the fraction of those shortest paths that pass through it.

Problem 3

Step 3: Edge Removal Strategy

a. Identify the edge with the absolute highest betweenness centrality score.

b. Remove this edge from the graph.

Problem 1

Step 4: Iterative Global Recalculation

a. Recalculate the betweenness for all remaining edges in the new topology.

b. Repeat Step 3 and 4 until the network is divided into the required number of sectors.

Step 5: Output Sectors

a. Identify the resulting disconnected sub-graphs as the final sectors.

Step 6: End

1.2 Statement of the Problem

Significant issues persist regarding the computational and practical execution of the GN Algorithm. Its standard implementation suffers from fundamental algorithmic flaws that severely limit its utility for large-scale, complex networks. Consequently, without integrating straightforward algorithmic solutions, the traditional GN method remains a computationally exhaustive and rigidly flawed algorithm. The core problem lies in transforming this limited algorithm into a fast, stable, and practical tool capable of robust network analysis.

The specific problems of this study are:

1. **The algorithm iteratively recalculates shortest paths after every single edge removal, which leads to an unfeasible time complexity for large-scale networks.**

   Calculating edge betweenness centrality for all edges requires a shortest-path search from every node in the network, creating an extreme computational bottleneck. Because removing an edge alters the shortest paths across the remaining topology, the betweenness values must be continuously re-evaluated at every iteration. This "recalculation paradox" imposes an unfeasible O(n^3) time complexity for sparse graphs, rendering the standard GN algorithm practically obsolete and incapable of processing modern large-scale networks efficiently without severe computational burden (Palacio-Niño & Berzal, 2024; Rustamaji et al., 2024; Godha & Prajapat, 2025).

Figure 1.1

Computational Complexity and Execution Time Scaling

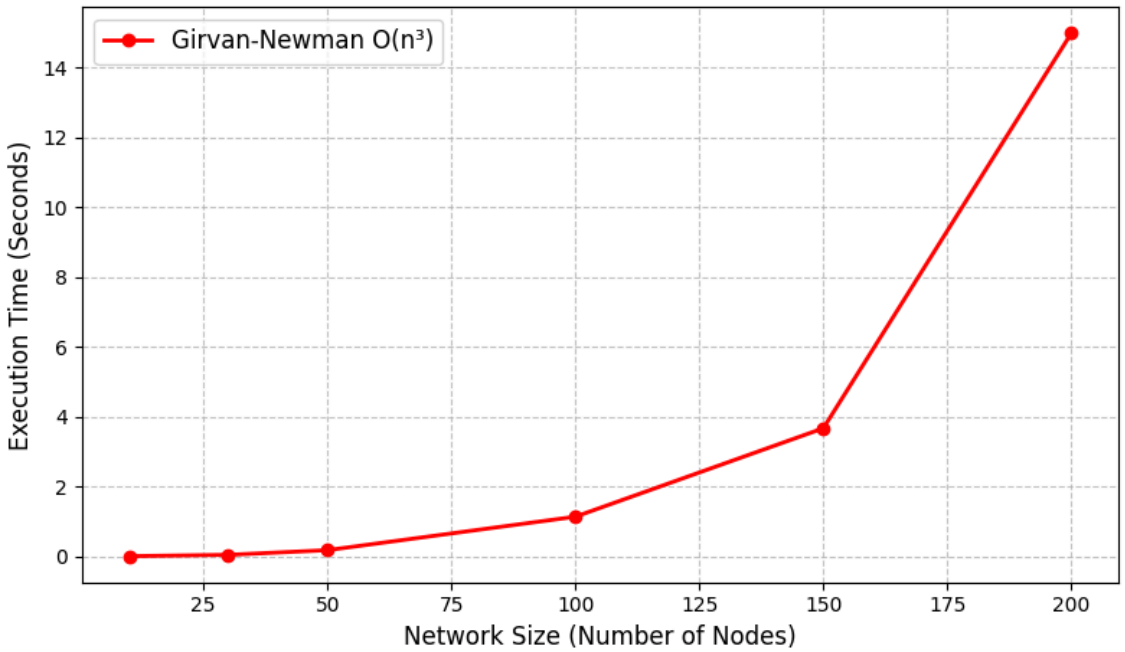

Figure 1.1 illustrates the exponential growth in execution time required by the Girvan-Newman algorithm as the number of nodes and edges in a network expands. Unlike alternative methods such as Louvain or Leiden which scale efficiently, the continuous recalculation of edge betweenness depicted in the chart highlights the algorithm's restrictive O(n^3) complexity. This exponential curve directly supports the problem statement by demonstrating the algorithm's prohibitive computational slowness and unsuitability for large datasets.

2. **The algorithm employs a greedy, top-down divisive approach lacking community size validation, which leads to the premature fragmentation of graphs and the erroneous classification of single isolated nodes as entire communities.**

The GN algorithm identifies the single edge with the highest betweenness and severs it without evaluating if a different sequence of removals might yield a more globally optimal partition. Once a split is performed, it is entirely irreversible. This rigid structural greediness and susceptibility to resolution constraints frequently lead to "early edge splits" that create meaningless "singleton" communities, where highly active individual nodes are snipped off rather than properly grouped, failing to accurately capture robust community boundaries (Pourabbasi et al., 2021; Shi, 2025; Bekkair et al., 2025).

Figure 1.2

Dendrogram of Premature Subgraph Fragmentation in Divisive Clustering

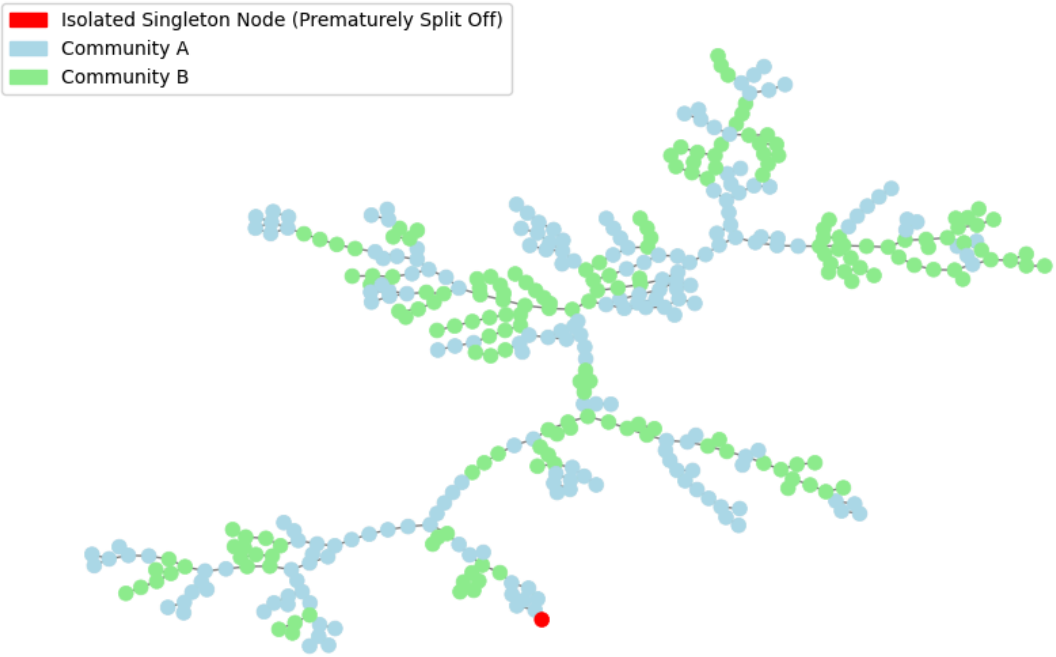

Figure 1.2 visually represents the irreversible, top-down edge removal process inherent to the GN method. It shows how the algorithm's greediness targets internal high-traffic edges prematurely, permanently separating single nodes from their natural structural groups before resolving actual community boundaries. This validates the problem statement by showcasing how the

algorithm's lack of a minimum size check inevitably yields suboptimal, fragmented partitions and meaningless singletons.

3. **The algorithm arbitrarily breaks ties when multiple edges share identical maximum betweenness scores, which leads to unstable, irreproducible partitions that lack graph-theoretical invariance.**

In many networks, particularly highly regular or symmetrical ones, multiple edges frequently tie for the exact same maximum betweenness centrality score. The standard implementation of the GN algorithm typically removes only one of these edges at a time, breaking the tie arbitrarily based on vertex labeling or the internal data ordering of the edge list. Without a deterministic tie-breaking mechanism, the algorithm's outputs can wildly diverge simply by reordering the input dataset, producing highly unstable and inconsistent community structures across different runs (Anuar et al., 2021; Pourabbasi et al., 2022; Bekkair et al., 2025).

Figure 1.3

Algorithmic Instability and Output Divergence Due to Arbitrary Tie-Breaking

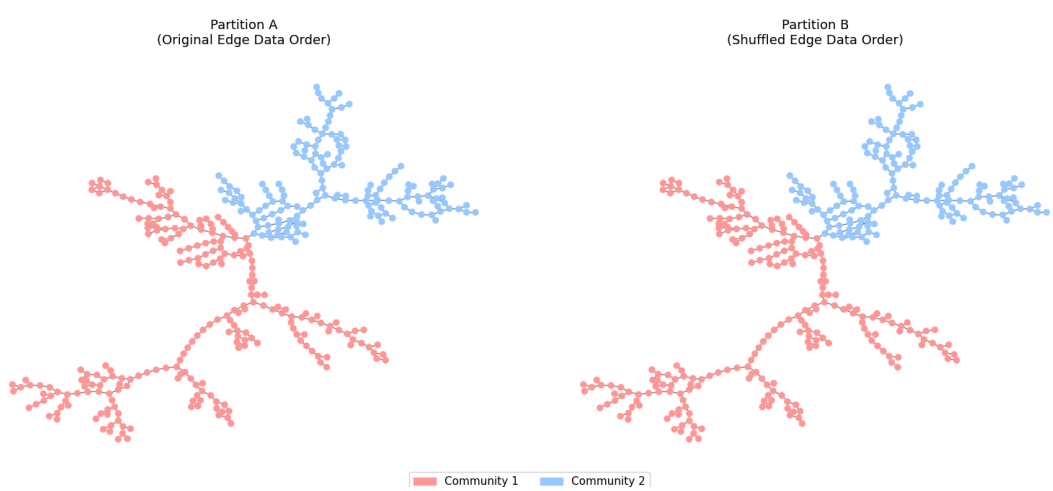

Figure 1.3 displays two completely different community detection results derived from identical starting networks. It demonstrates how varying the initial

internal data ordering forces the algorithm to arbitrarily cut different edges when encountering a tie in betweenness scores. This visual directly supports the problem statement by illustrating the algorithm's irreproducibility and critical lack of structural stability.

1.3 Objective of the Study

The main objective of this study is to enhance the Girvan-Newman (GN) algorithm to achieve computationally efficient and hydraulically stable water network sectorization in urban infrastructure.

The specific objectives are to:

1. Implement a betweenness approximation technique (such as network traffic sampling) to reduce the computational latency associated with the iterative global re-calculation of shortest paths, resolving the algorithm's unfeasible O(n^3) time complexity in high-dimensional graphs.
2. Optimize the top-down divisive logic by integrating a minimum community size validation check into the edge removal process, thereby preventing premature graph fragmentation and eliminating the erroneous generation of meaningless single-node communities.
3. Modify the algorithm's edge selection criteria to systematically handle ties by simultaneously removing all edges sharing the identical maximum betweenness score, ensuring that the resulting partitions are completely stable, reproducible, and graph-theoretical invariant.

1.4 Significance of the Study

This research addresses the critical intersection of graph theory and civil infrastructure, providing a computational framework that balances algorithmic efficiency with physical reality.

Water Utility Providers. Allows for faster, cheaper planning of sectorization to reduce water loss (Non-Revenue Water).

Urban Engineers. Provides a tool that respects the laws of physics (pressure/flow) while offering the benefits of graph-based partitioning.

Future Researchers. Advances the field of Constrained Graph Partitioning by applying a hybrid metaheuristic approach to real-world infrastructure.

1.5 Scope and Limitations

This section defines the operational boundaries and the specific technical constraints under which this research will be conducted.

The primary focus of this research is the algorithmic modification and enhancement of the Girvan-Newman (GN) algorithm. It specifically includes the development of a Hydraulically-Aware variant of the GN algorithm. This replaces traditional unit-weight edges with weights derived from physical pipe resistance, such as Hazen-Williams or Darcy-Weisbach coefficients, alongside nodal demand data. Another core component is the implementation of a Local Betweenness Centrality approximation designed to bypass the global recalculation bottleneck, directly targeting a reduction in processing time for large-scale graphs. The algorithm will be tested and validated using three internationally recognized benchmark water distribution datasets: Hanoi for initial algorithmic verification, C-Town for testing medium-scale sectorization accuracy, and L-Town for measuring computational scalability on complex urban grids. Performance will be evaluated based on execution time in CPU seconds, a Hydraulic Resilience Index using nodal pressure consistency, and the Modularity Score to measure the strength of the resulting community structure.

While this study aims to provide a robust solution for infrastructure partitioning, several constraints apply. The research focuses on steady-state network conditions and will not account for transient hydraulic events like water hammers or second-by-second fluctuations in consumer behavior. The algorithm assumes a fixed, static network

topology and does not account for real-time valve operations or dynamic rerouting during emergency repairs. And in addition to that, this research is strictly computational and simulation-based; it does not involve the physical installation of sensors or valves in a live municipal environment. The accuracy of the enhancement is dependent on the mathematical models provided by the EPANET 2.2 engine and the WNTR Python library. Finally, the study is limited to the physical parameters provided in the benchmark files, and external socio-economic factors like population growth or zoning laws are considered outside the scope of this algorithmic optimization.

1.6 Definition of Terms

This section aims to define and clarify the key terminology associated with the study, arranged alphabetically.

Betweenness Centrality – A graph-theoretic metric that identifies the importance of an edge or node based on the frequency with which it lies on the shortest paths between all node pairs within the network topology.

Boundary Edge – A connection, specifically a pipe in a water distribution network, that bridges two different partitions and is typically where isolation valves or bulk flow meters are installed during sectorization.

Computational Complexity (O notation) – A mathematical representation used in computer science to describe the limiting behavior of an algorithm's execution time or space requirements as the input size increases.

District Metered Area (DMA) – A discrete, isolated zone of a water distribution network where the total water entering and leaving the area is metered to allow for precise leakage analysis and pressure control.

EPANET/WNTR – A specialized public-domain hydraulic simulation engine and its corresponding Python-based research library used to model the physical behavior of pressurized water systems.

Graph Partitioning – The computational process of dividing the set of vertices of a graph into smaller, non-overlapping subsets while minimizing the number of edges that connect different subsets.

Hydraulic Awareness – The integration of physical system properties, such as pressure gradients, flow resistance, and nodal demand, into a mathematical algorithm to ensure physical consistency.

Local Betweenness Recalculation – An algorithmic optimization that updates centrality scores only for a specific neighborhood affected by an edge removal, rather than performing a global re-scan of the entire network.

Modularity (Q score) – A scalar value used to measure the strength of the division of a network into clusters, where high modularity indicates dense connections between nodes within clusters and sparse connections between nodes in different clusters.

Non-Revenue Water (NRW) – The difference between the volume of water put into a distribution system and the volume of water billed to customers, representing losses due to leaks, theft, or metering inaccuracies.

Redundant Looped Network – A structural configuration typical of urban infrastructure where multiple pathways exist between any two nodes to ensure system reliability and resilience in the event of a localized failure.

Step-wise Recalculation Bottleneck – The computational delay caused by the requirement of certain hierarchical algorithms to re-compute global metrics after every iterative change to the graph topology.

Water Network Sectorization (WNS) – The strategic process of dividing a large-scale water grid into smaller, independent zones to optimize pressure management, improve leak detection, and enhance operational resilience.

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This is the full citation for the paper cited in your text as "Godha & Prajapat, 2025." You may want to update the in-text citation to "Sakle et al., 2025" to perfectly match APA 7th standards).

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